Quality of services (QoS)

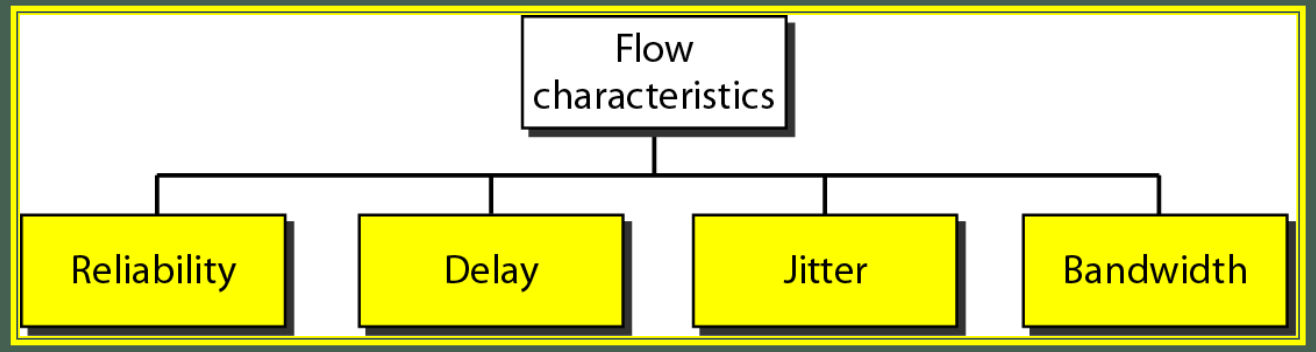
Content

- Introduction
- Techniques to Improve QOS
- Integrated Services

Introduction

- *Quality of service (QoS) is an internetworking issue that has been discussed more than defined.*
- *We can informally define quality of service as something a flow seeks to attain.*

Flow characteristics



Reliability

- Reliability is a characteristic that a flow needs. Lack of reliability means losing a packet or acknowledgment, which entails retransmission.
- The sensitivity of application programs to reliability is not the same.
- For example, it is more important that electronic mail, file transfer, and Internet access have reliable transmissions than telephony or audio conferencing.

Delay

- Source-to-destination delay is another flow characteristic.
- Again applications can tolerate delay in different degrees.
- In this case, telephony, audio conferencing, video conferencing, and remote log-in need minimum delay, while delay in file transfer or e-mail is less important.

Jitter

- Jitter is the variation in delay for packets belonging to the same flow.
- For example, if four packets depart at times 0, 1, 2, 3 and arrive at 20, 21, 22, 23, all have the same delay, 20 units of time.
- On the other hand, if the above four packets arrive at 21, 23, 21, and 28, they will have different delays: 21,22, 19, and 24.
- For applications such as audio and video, the first case is completely acceptable; the second case is not.

Jitter

- For these applications, it does not matter if the packets arrive with a short or long delay as long as the delay is the same for all packets. For this application, the second case is not acceptable.
- Jitter is defined as the variation in the packet delay. High jitter means the difference between delays is large; low jitter means the variation is small.

Bandwidth

- Different applications need different bandwidths.
- In video conferencing we need to send millions of bits per second to refresh a color screen while the total number of bits in an e-mail may not reach even a million.

TECHNIQUES TO IMPROVE QoS

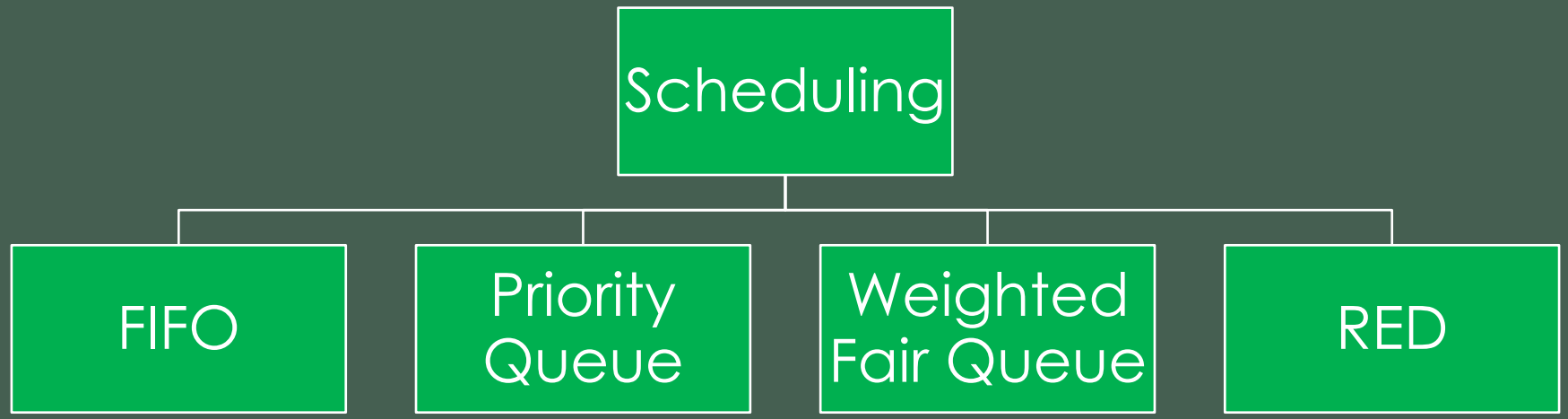
Four common methods:

- Scheduling
- Traffic shaping
- Admission control
- Resource reservation

Scheduling

- Packets from different flows arrive at a switch or router for processing.
- A good scheduling technique treats the different flows in a fair and appropriate manner.
- Several scheduling techniques are designed to improve the quality of service.

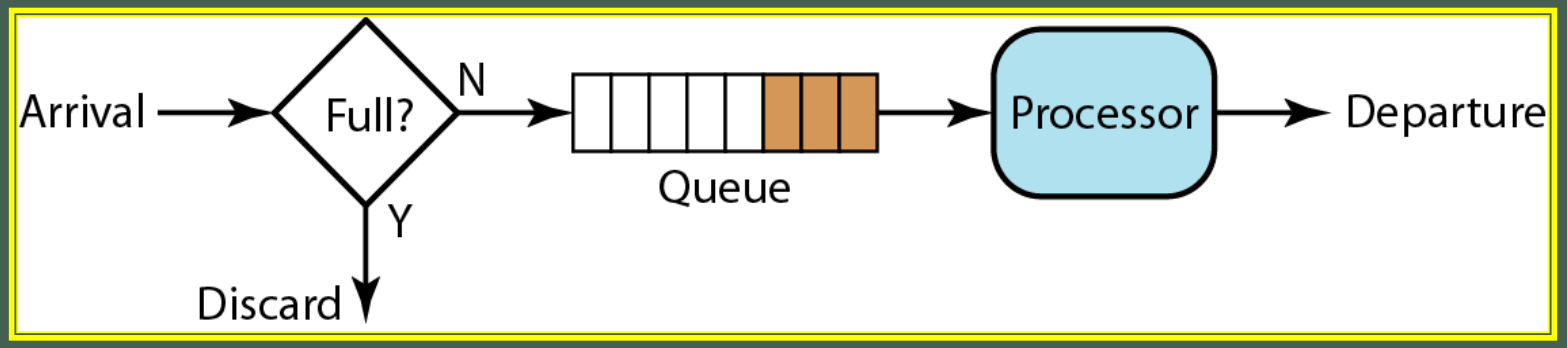
Scheduling



FIFO Queuing

- In first-in, first-out (FIFO) queuing, packets wait in a buffer (queue) until the node (router or switch) is ready to process them.
- If the average arrival rate is higher than the average processing rate, the queue will fill up and new packets will be discarded.

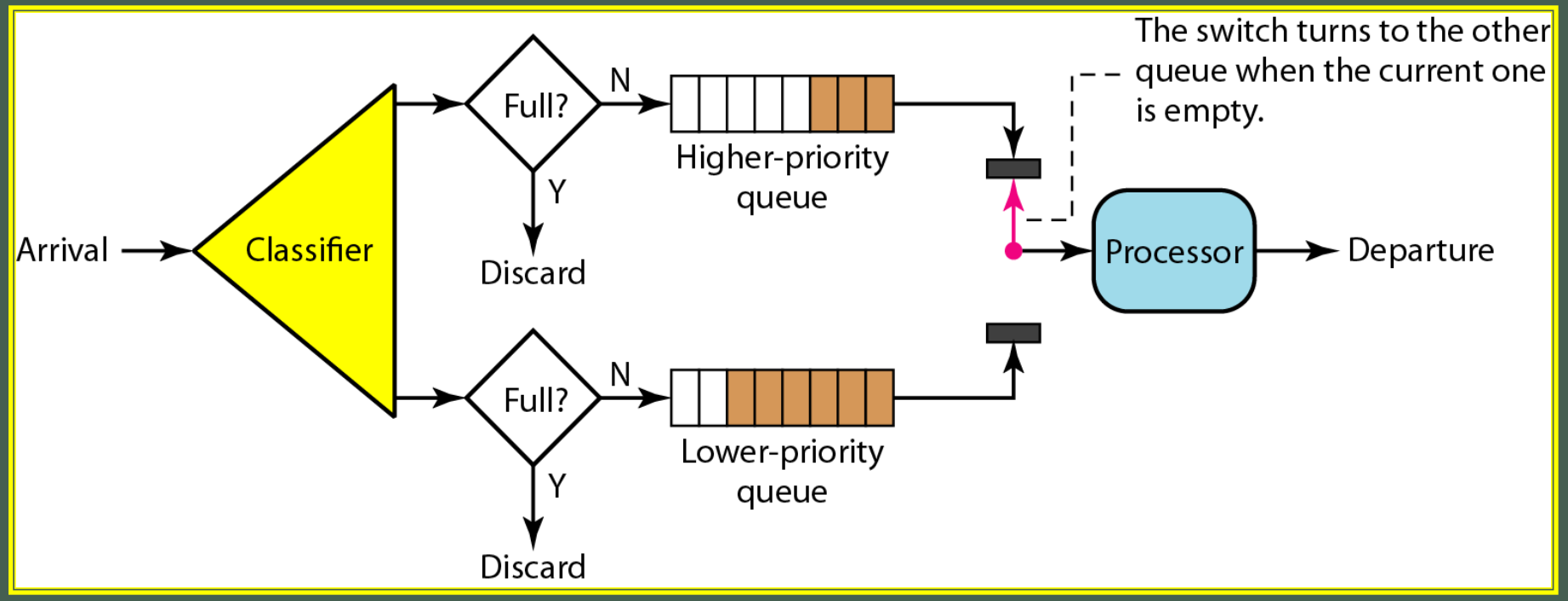
FIFO queue



Priority Queuing

- In priority queuing, packets are first assigned to a priority class. Each priority class has its own queue.
- The packets in the highest-priority queue are processed first. Packets in the lowest-priority queue are processed last.
- A priority queue can provide better QoS than the FIFO queue because higher priority traffic, such as multimedia, can reach the destination with less delay.

Priority queuing



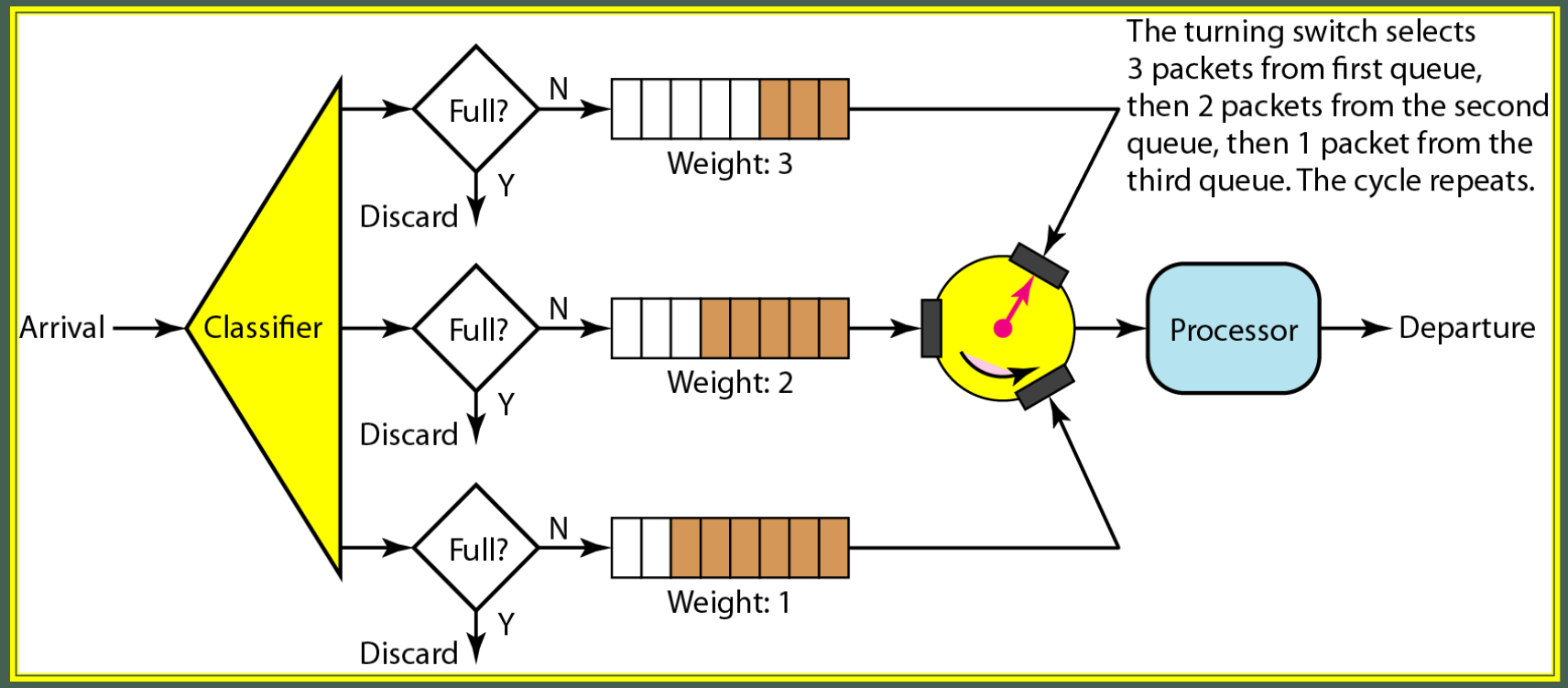
Weighted Fair Queuing

- A better scheduling method is weighted fair queuing. In this technique, the packets are still assigned to different classes and admitted to different queues.
- The queues, however, are weighted based on the priority of the queues; higher priority means a higher weight.
- The system processes packets in each queue in a round-robin fashion with the number of packets selected from each queue based on the corresponding weight.

Contd…

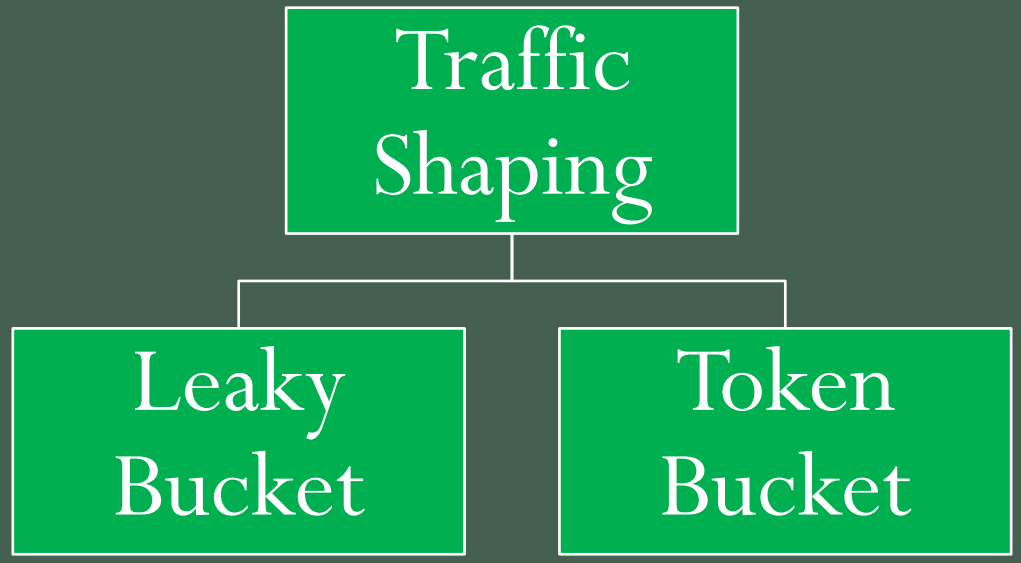
- For example, if the weights are 3, 2, and 1, three packets are processed from the first queue, two from the second queue, and one from the third queue.
- If the system does not impose priority on the classes, all weights can be equal In this way, we have fair queuing with priority.

Weighted fair queuing

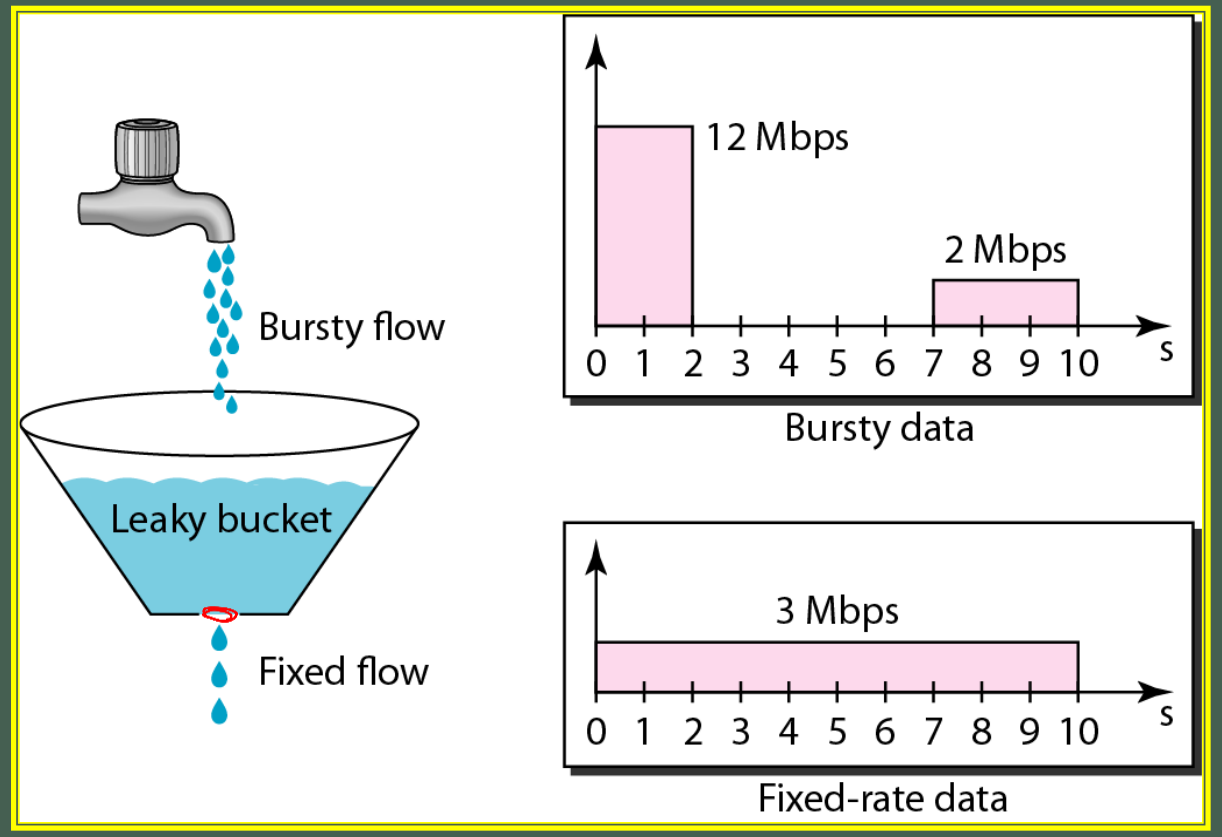


Traffic Shaping

Traffic shaping is a mechanism to control the amount and the rate of the traffic sent to the network.



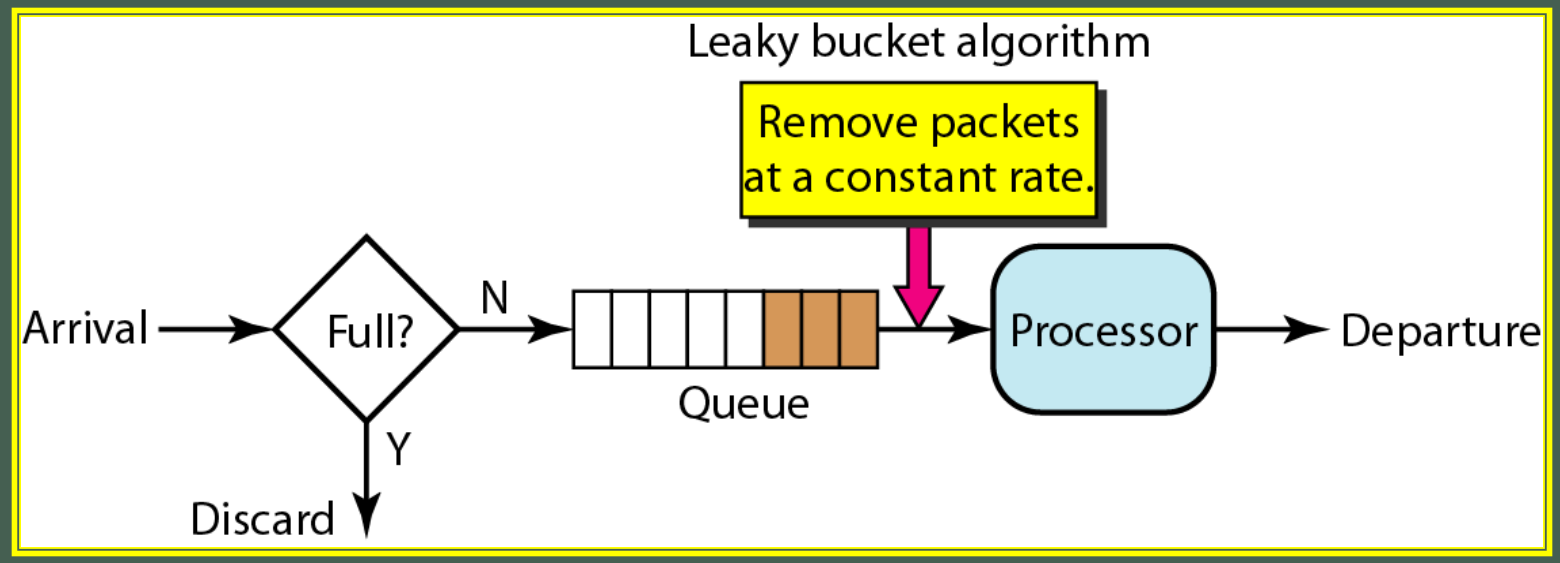
Leaky bucket



Leaky Bucket

- If a bucket has a small hole at the bottom, the water leaks from the bucket at a constant rate as long as there is water in the bucket.
- The rate at which the water leaks does not depend on the rate at which the water is input to the bucket unless the bucket is empty.
- The input rate can vary, but the output rate remains constant.
- Similarly, in networking, a technique called leaky bucket can smooth out bursty traffic. Bursty chunks are stored in the bucket and sent out at an average rate.

Leaky bucket implementation



Note

A leaky bucket algorithm shapes bursty traffic into fixed-rate traffic by averaging the data rate. It may drop the packets if the bucket is full.

Token Bucket

- The leaky bucket is very restrictive. It does not credit an idle host. For example, if a host is not sending for a while, its bucket becomes empty.
- Now if the host has bursty data, the leaky bucket allows only an average rate. The time when the host was idle is not taken into account.
- On the other hand, the token bucket algorithm allows idle hosts to accumulate credit for the future in the form of tokens.
- For each tick of the clock, the system sends n tokens to the bucket.

Contd…

- The system removes one token for every cell (or byte) of data sent.
- For example, if n is 100 and the host is idle for 100 ticks, the bucket collects 10,000 tokens.
- Now the host can consume all these tokens in one tick with 10,000 cells, or the host takes 1000 ticks with 10 cells per tick.
- In other words, the host can send bursty data as long as the bucket is not empty

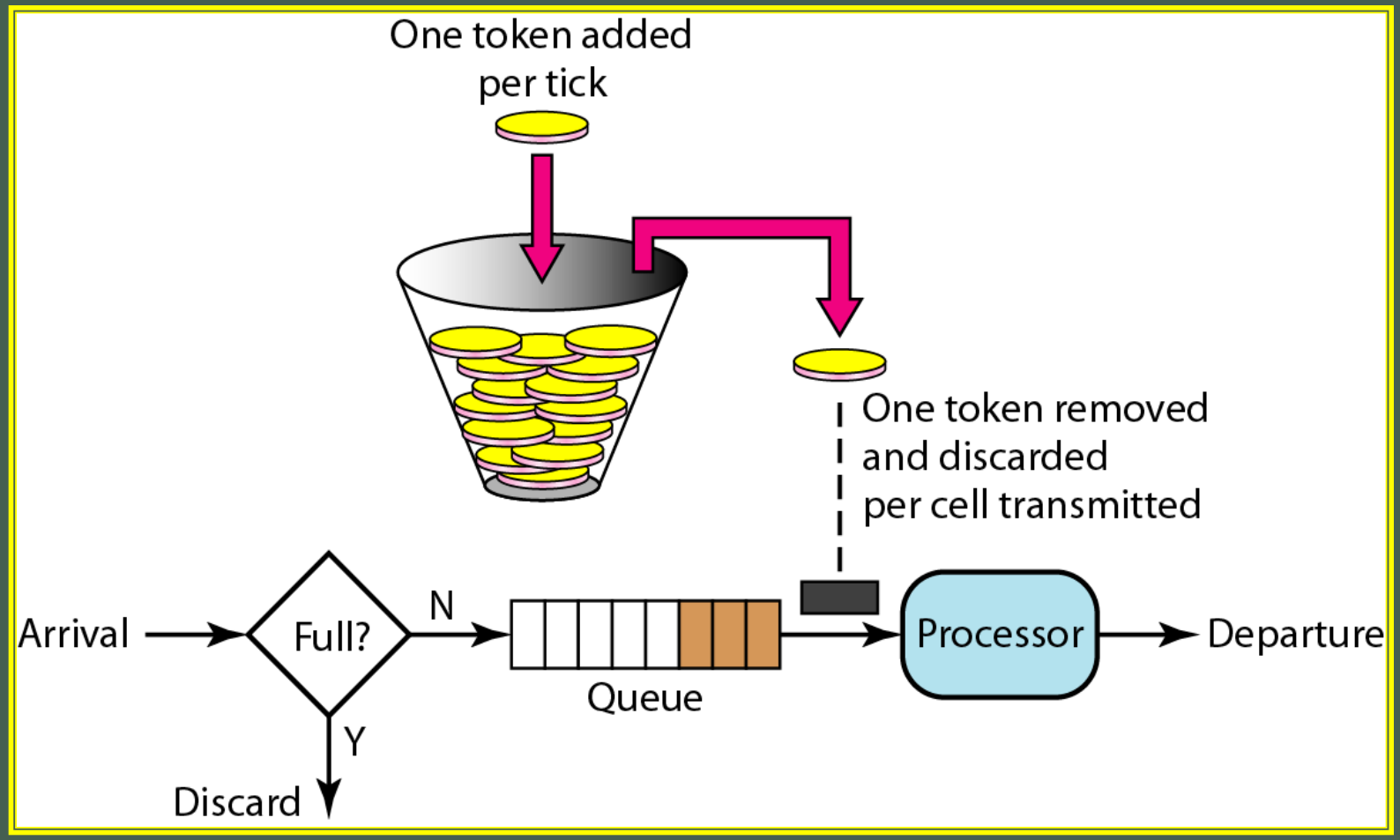
Contd…

- The token bucket can easily be implemented with a counter.
- The token is initialized to zero. Each time a token is added, the counter is incremented by 1.
- Each time a unit of data is sent, the counter is decremented by 1.
- When the counter is zero, the host cannot send data.

Note

The token bucket allows bursty traffic at a regulated maximum rate.

Token bucket



Resource Reservation and Admission Control

- Buffer, CPU Time, Bandwidth are the resources that can be reserved for particular flows for particular time to maintain the QoS.
- Mechanism used by routers to accept or reject flows based on flow specifications is what we call Admission Control.

INTEGRATED SERVICES

- Two models have been designed to provide quality of service in the Internet: Integrated Services and Differentiated Services.
- Both models emphasize the use of quality of service at the network layer (IP), although the model can also be used in other layers such as the dataline.
- Integrated Services is a flow based QoS model designed for IP.

Signaling

- IP is a connectionless, datagram, packet-switching protocol.
- How can we implement a flow-based model over a connectionless protocol?
- The solution is a signaling protocol to run over IP that provides the signaling mechanism for making a reservation.
- This protocol is called Resource Reservation Protocol (RSVP)

Flow Specification

- When a source makes a reservation, it needs to define a flow specification.
- A flow specification has two parts: Rspec (resource specification) and Tspec (traffic specification).
- Rspec defines the resource that the flow needs to reserve (buffer, bandwidth, etc.). Tspec defines the traffic characterization of the flow.

Admission

- After a router receives the flow specification from an application, it decides to admit or deny the service.
- The decision is based on the previous commitments of the router and the current availability of the resource.

Service Classes

- Two classes of services have been defined for Integrated Services:
  - Guaranteed service
  - Controlled-load service

Guaranteed Service Class

- This type of service is designed for real-time traffic that needs a guaranteed minimum end-to-end delay.
- The end-to-end delay is the sum of the delays in the routers, the propagation delay in the media, and the setup mechanism.
- Only the first, the sum of the delays in the routers, can be guaranteed by the router.
- This type of service guarantees that the packets will arrive within a certain delivery time and are not discarded if flow traffic stays within the boundary of Tspec.

Controlled-Load Service Class

- This type of service is designed for applications that can accept some delays, but are sensitive to an overloaded network and to the danger of losing packets.
- Good examples of these types of applications are file transfer, e-mail, and Internet access.
- The controlled load service is a qualitative type of service in that the application requests the possibility of low-loss or no-loss packets.

RSVP

- In the Integrated Services model, an application program needs resource reservation. The IntServ model, the resource reservation is for a flow.
- A virtual-circuit network needs a signaling system to set up the virtual circuit before data traffic can start.
- The Resource Reservation Protocol (RSVP) is a signaling protocol to help IP create a flow and consequently make a resource reservation.

Multicast Trees

- RSVP is different from some other signaling systems we have seen before in that it is a signaling system designed for multicasting.
- RSVP can be also used for unicasting because unicasting is just a special case of multicasting with only one member in the multicast group.
- The reason for this design is to enable RSVP to provide resource reservations for all kinds of traffic including multimedia which often uses multicasting.

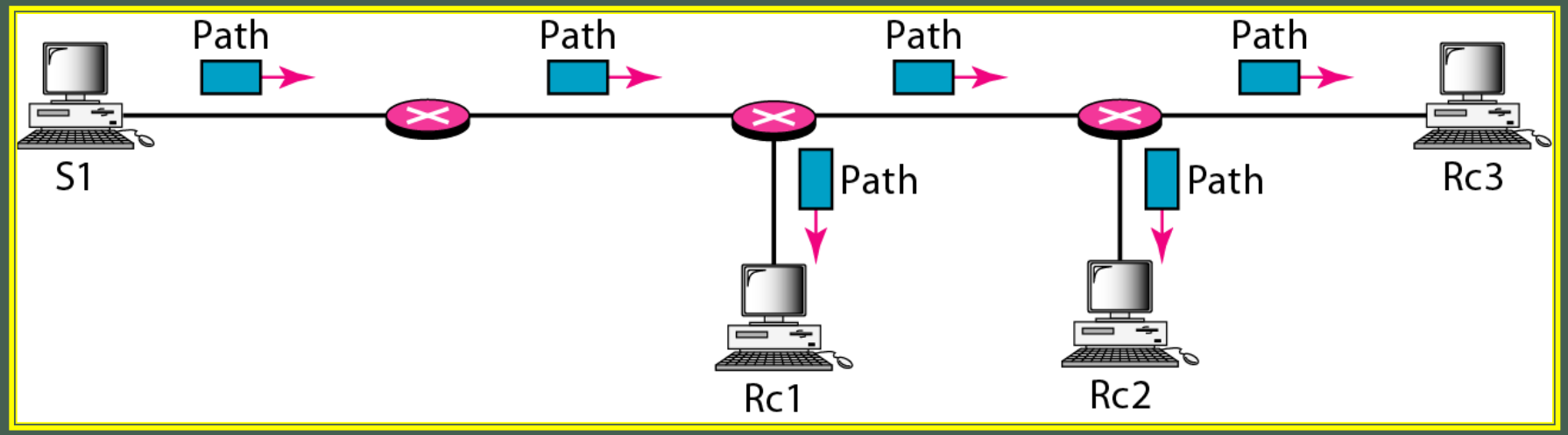
Receiver-Based Reservation

- In RSVP, the receivers, not the sender, make the reservation. This strategy matches the other multicasting protocols.
- For example, in multicast routing protocols, the receivers, not the sender, make a decision to join or leave a multicast group.

RSVP Messages

- RSVP has several types of messages: Path and Resv.
- Path Messages: Recall that the receivers in a flow make the reservation in RSVP.
- The receivers do not know the path traveled by packets before the reservation is made. The path is needed for the reservation.
- To solve the problem, RSVP uses Path messages. A Path message travels from the sender and reaches all receivers in the multicast path.
- On the way, a Path message stores the necessary information for the receivers.
- A Path message is sent in a multicast environment; a new message is created when the path diverges.

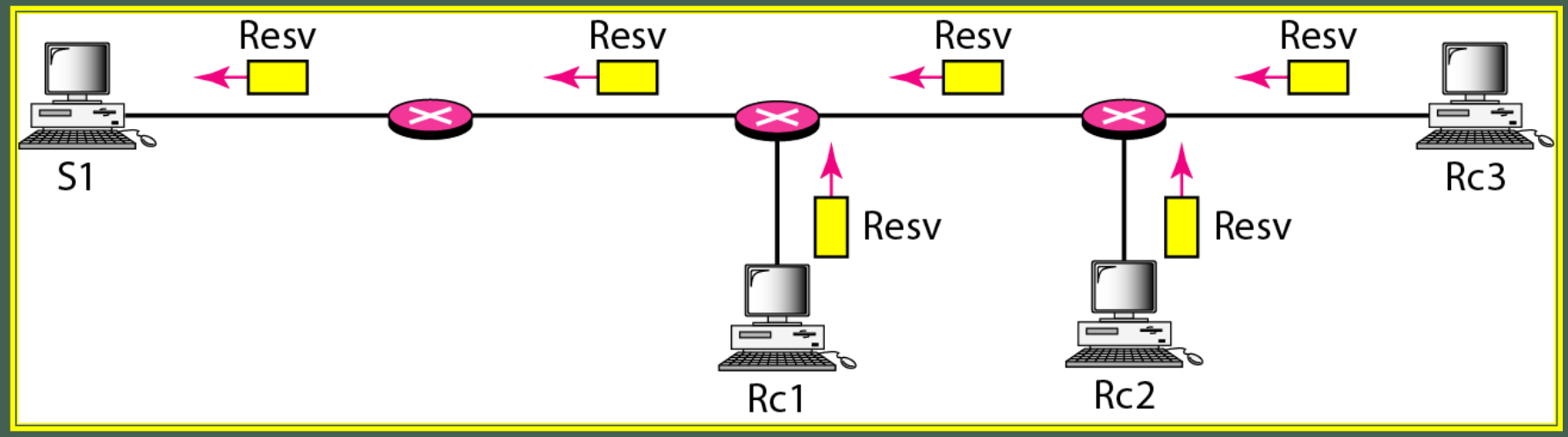
Path messages



Contd…

- Resv Messages: After a receiver has received a Path message, it sends a Resv message.
- The Resv message travels toward the sender (upstream) and makes a resource reservation on the routers that support RSVP.
- If a router does not support RSVP on the path, it routes the packet based on the best-effort delivery methods.

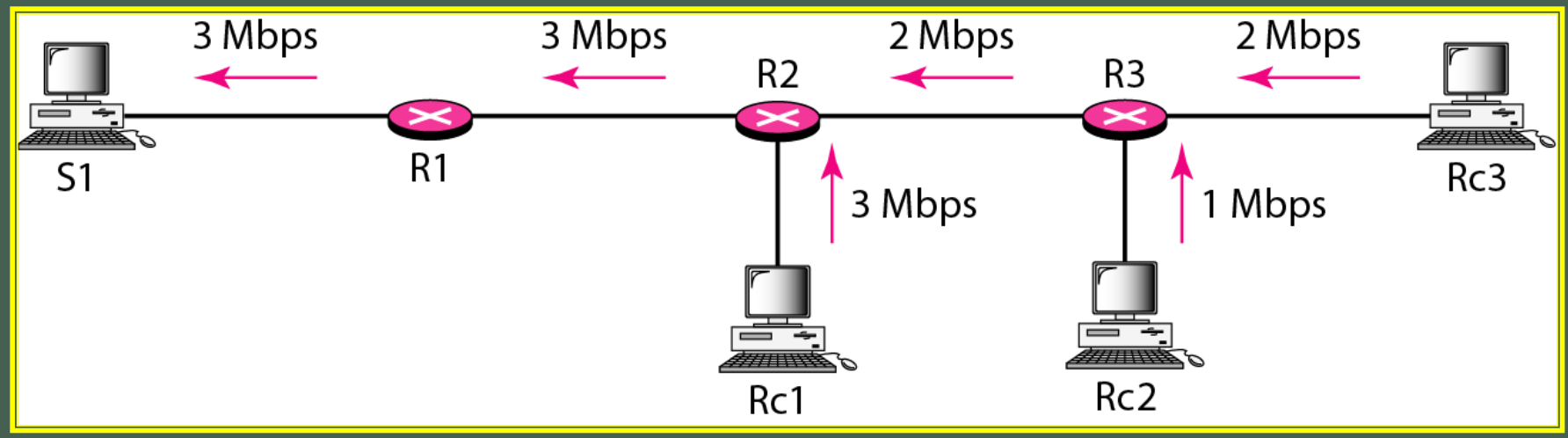
Resv messages



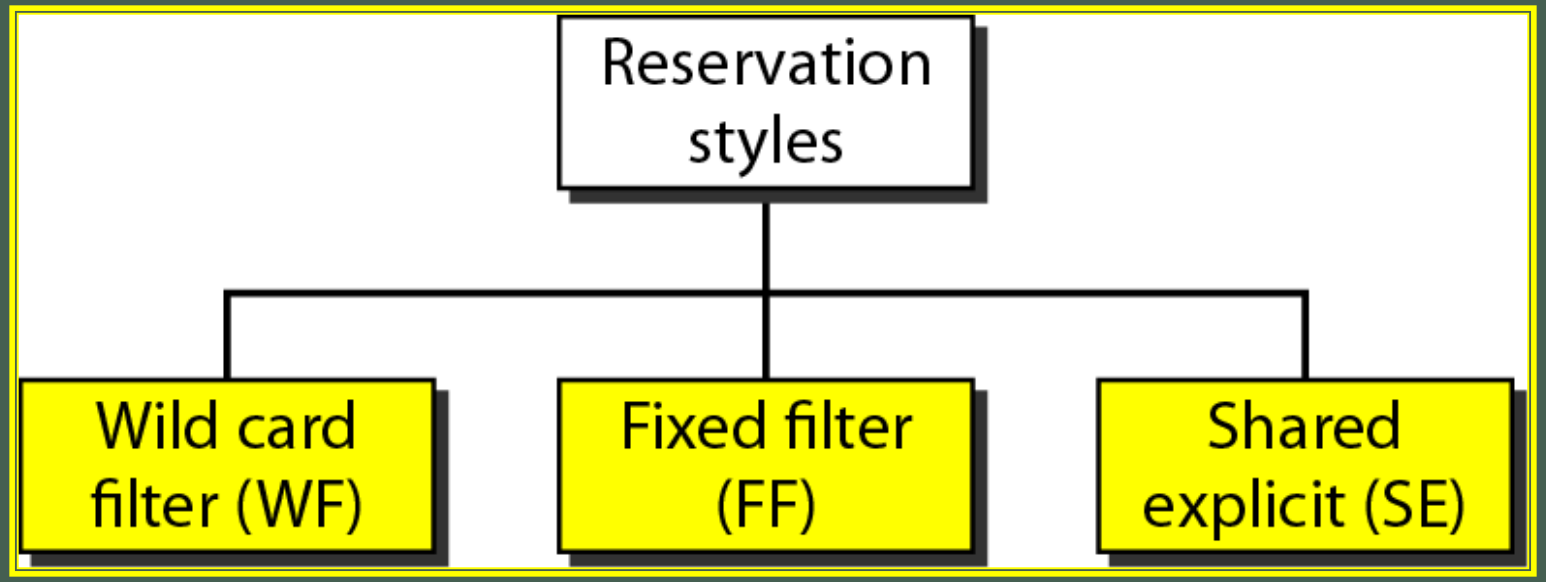
Reservation Merging

- In RSVP, the resources are not reserved for each receiver in a flow; the reservation is merged.
- RC3 requests a 2-Mbps bandwidth while RC2 requests a 1-Mbps bandwidth. Router R3, which needs to make a bandwidth reservation, merges the two requests.
- The reservation is made for 2 Mbps, the larger of the two, because a 2-Mbps input reservation can handle both requests.
- The same situation is true for R2.

Reservation merging



Reservation styles



Contd…

- Wild Card Filter Style: In this style, the router creates a single reservation for all senders. The reservation is based on the largest request. This type of style is used when the flows from different senders do not occur at the same time.
- Fixed Filter Style: In this style, the router creates a distinct reservation for each flow. This means that if there are n flows, n different reservations are made. This type ofstyle is used when there is a high probability that flows from different senders will occur at the same time.
- Shared Explicit Style: In this style, the router creates a single reservation which can be shared by a set of flows